

Quantum Computing Standards and Certifications Research

International Standards Bodies

1. ISO/IEC Standards

- **ISO/IEC 4879:2024:** Information technology — Quantum computing — Vocabulary
- Defines terms commonly used in quantum computing
- Covers quantum physics background, quantum information, quantum processing, quantum technologies
- Applicable to all types of organizations for exchanging quantum computing concepts

2. IEEE Standards

- **IEEE Standardization Roadmap on Quantum Applications**
- Focus on application-readiness and stakeholder guidance
- Links business problems to quantum technologies
- Addresses standardization needs across supply chain, reliability, international distribution

3. CEN-CENELEC

- **Standardization Roadmap on Quantum Technologies**
- European focus on quantum technology standards
- Creates common language for quantum computing field

Existing Quantum Computing Certifications

1. IBM Certified Associate Developer - Quantum Computation using Qiskit v0.2X

- **Provider:** IBM
- **Focus:** Qiskit framework for quantum computing
- **Level:** Associate Developer
- **Prerequisites:** Familiarity with quantum circuits, Pauli matrices, quantum state modeling
- **Cost:** \$200 per exam
- **Format:** Online, self-paced

2. Microsoft Azure Quantum Developer Certification

- **Provider:** Microsoft
- **Focus:** Q# programming language and Azure Quantum
- **Level:** Beginner-friendly
- **Prerequisites:** Basic Azure knowledge, linear algebra, Visual Studio Code
- **Cost:** Free
- **Format:** Online, self-paced

3. Professional Certificate in Quantum Computing for Everyone

- **Provider:** University of Chicago via edX
- **Focus:** Core principles and business applications
- **Level:** Beginner with programming background
- **Duration:** 3 months
- **Cost:** \$358
- **Format:** Online, self-paced

4. Quantum Computing Fundamentals

- **Provider:** MIT xPRO
- **Focus:** Business and technical implications
- **Level:** Professional/Business-oriented
- **Duration:** 4 weeks, 4-6 hours per week
- **Cost:** \$2,419
- **Format:** Online, self-paced

5. Graduate Certificate in Quantum Computing

- **Provider:** University of Maryland
- **Focus:** Advanced knowledge and hands-on training
- **Level:** Graduate
- **Duration:** One year (12 credits)
- **Cost:** \$4,146
- **Format:** In-person
- **Prerequisites:** Bachelor's degree, 3.0 GPA, linear algebra, programming proficiency

Other Certification Bodies

6. Certified Quantum Security Professional (CQSP)

- **Provider:** SISA InfoSec
- **Focus:** Quantum security and post-quantum cryptography
- **Target:** Cybersecurity professionals

7. Certified Quantum Practitioner (CQP)

- **Provider:** International Institute of Quantum Computing (i2QC) / Tonex
- **Focus:** Navigating quantum technologies

- **Target:** Professionals

8. Quantum Standards Group (QSG)

- **Provider:** EITCI (European IT Certification Institute)
- **Focus:** Industry specifications and standards
- **Target:** International experts

Key Insights for Curriculum Development

Foundation Level (3 Days) - "Quantum for Everyone"

Should cover: 1. Quantum computing fundamentals (qubits, superposition, entanglement) 2. Quantum vs classical computing 3. Business applications and use cases 4. Post-quantum cryptography basics 5. Quantum threat landscape 6. No-code quantum tools (IBM Quantum Composer, Azure Quantum)

Practitioner Level (5 Days)

Should cover: 1. Quantum algorithms and circuits 2. Quantum programming (Qiskit or Q#) 3. Quantum hardware and architectures 4. Advanced PQC implementation 5. Sector-specific applications (telecom, banking, healthcare, retail) 6. Hands-on quantum development

Market Context

- Quantum technology market estimated to reach \$106 billion by 2040 (McKinsey)
- 55% of quantum jobs require bachelor's degree or less (Chicago Quantum Exchange study)
- Growing demand for quantum-skilled professionals across industries

Standards Timeline (EU Roadmap)

- **2026:** National strategies and awareness campaigns established

- **2030:** Critical infrastructure and high-risk use cases protected with PQC
- **2035:** Broad adoption across all sectors and lower-risk systems completed